

Causal Inference and ML: Modeling Coping Strategy Improvement

Estimating the influence of the socioeconomic environment on individual coping strategies

Github Repository:

github.com/marthab-oss/Modeling-Coping-Strategy-Improvement-via-Causal-Inference

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Examine the influence of different variables on the individual Coping Skill:

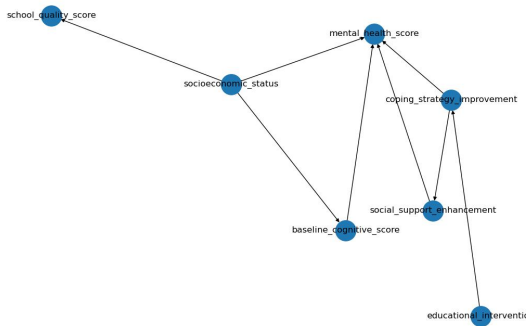
- **Outcome:** Coping Strategy Improvement
- **Treatment:** Educational Intervention
- **Covariables:**
 - social support enhancement
 - mental health score
 - socioeconomic status
 - school quality score
 - baseline cognitive score
 - random noise

Data Structure - Dataset Excerpt

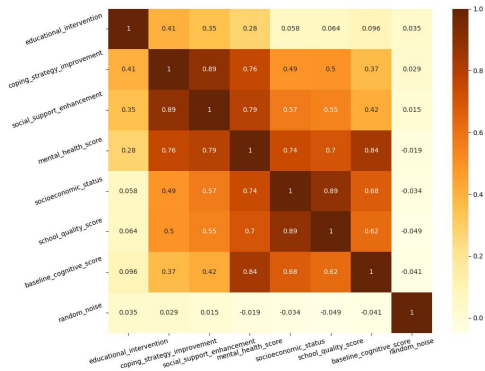
educational_intervention	coping_strategy_improvement	social_support_enhancement	mental_health_score	socioeconomic_status	school_quality_score	baseline_cognitive_score	random_noise
0	3.299	4.447	140.703	0.497	82.678	118.961	0.682
1	4.974	8.696	142.483	-0.138	81.532	107.864	0.126
0	3.646	7.473	138.472	0.648	74.587	107.073	-0.222
0	6.858	9.413	163.341	1.523	89.436	108.761	2.047
0	7.256	11.372	153.809	-0.234	70.524	104.641	-0.675
0	8.030	13.165	162.330	-0.234	71.178	101.593	-0.403
1	8.973	12.010	182.788	1.579	85.264	124.744	1.992
1	7.896	10.558	171.341	0.767	77.684	114.026	-0.832
1	3.327	6.779	139.325	-0.469	71.438	105.801	-0.550
1	6.106	9.725	156.019	0.543	76.017	100.073	-0.147
0	3.389	6.297	135.484	-0.463	72.374	108.540	0.841
0	3.586	6.279	128.820	-0.466	71.073	97.319	0.208
0	4.956	8.337	162.221	0.242	75.980	123.172	-1.293
0	2.195	1.886	89.753	-1.913	56.064	73.975	-0.532
0	0.179	2.410	103.319	-1.725	58.894	100.110	-0.607
0	5.669	9.725	143.092	-0.562	73.523	96.356	-0.077
1	3.938	5.744	109.827	-1.013	68.901	83.358	0.426
1	7.023	10.333	148.110	0.314	73.604	98.304	0.418
1	8.701	9.950	138.848	-0.908	68.133	87.716	-1.776
1	3.801	6.272	119.439	-1.412	66.707	90.119	1.064
1	3.932	7.938	157.370	1.466	80.048	119.885	0.253
0	4.428	6.926	125.519	-0.226	75.367	92.005	1.385

Data Structure - Relationship between Variables

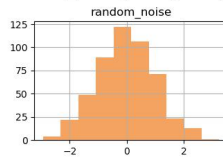
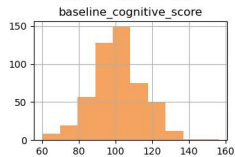
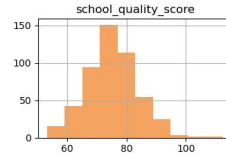
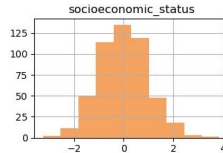
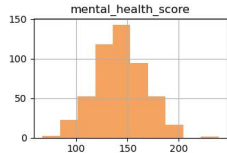
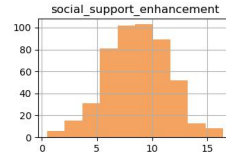
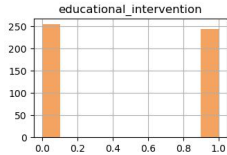
Statistical DAG



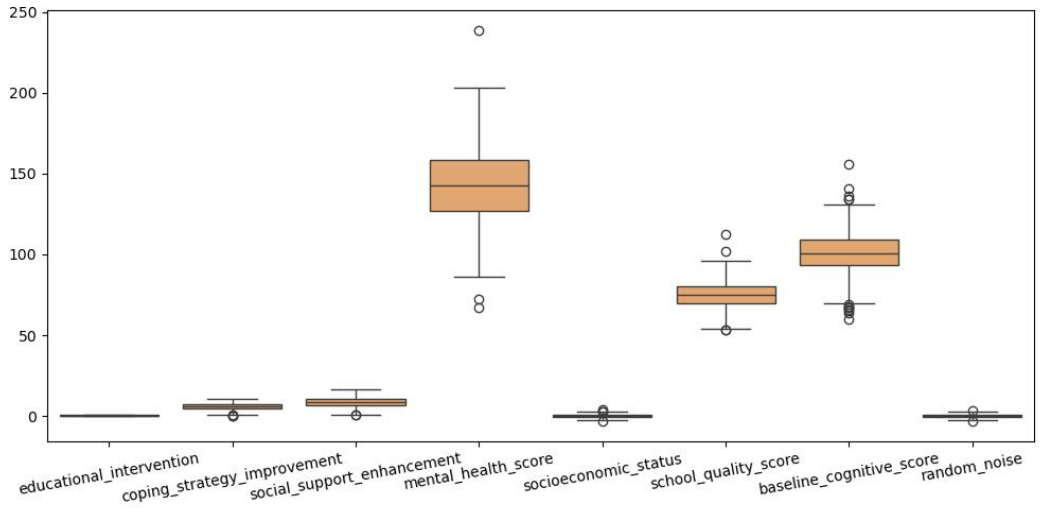
Correlation Matrix



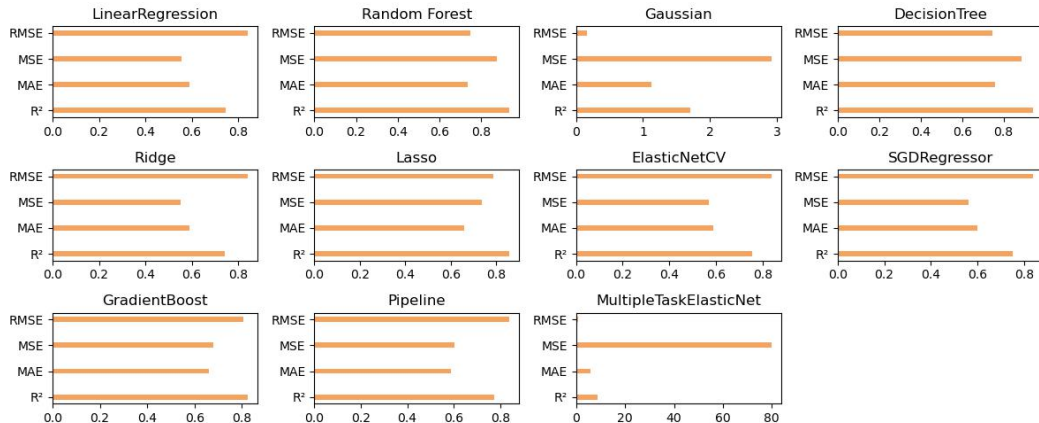
Data Structure - Visualization of Occurences



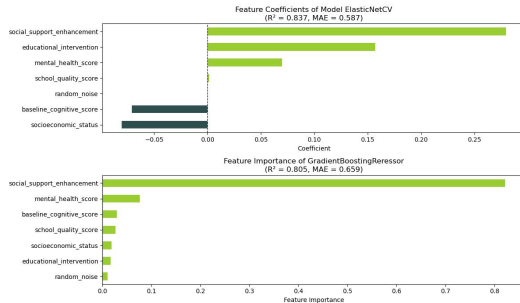
Data Structure - Outliers



Regression Models - Performance



Regression Models - Feature Importance



Causal DAG values

educational_intervention	coping_strategy_improvement	social_support_enhancement	mental_health_score	socioeconomic_status	school_quality_score	baseline_cognitive_score	random_noise
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.403	0.0	0.579	0.0	0.0	0.0	0.0	0.0
1.758	0.0	0.0	0.0	0.0	0.0	0.079	0.0
0.0	2.958	2.624	0.0	1.487	0.0	1.015	0.0
0.0	-0.075	0.083	0.0	0.0	0.083	0.013	0.0
-2.01	0.0	1.181	0.0	0.0	0.0	0.294	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

T,S,X-Learners and Causal Forests

